

Trend Following in Strategic Asset Allocation: A Long-Horizon Analysis and Retail-Oriented Implementation

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Abstract

Traditional portfolio construction frameworks rely on static asset allocation and cross-asset diversification to manage risk and improve long-term outcomes. Well-known approaches such as the All-Weather Portfolio and the Permanent Portfolio aim to reduce drawdowns and volatility through fixed exposures to multiple asset classes.

These frameworks have demonstrated robustness across a wide range of economic environments and remain a cornerstone of long-term portfolio construction. At the same time, recent literature has highlighted that portfolio risk is not solely determined by asset class composition, but also by the **time-varying nature of market regimes and return dynamics**.

This paper investigates the role of **trend following as a structural component of strategic asset allocation**, rather than as a standalone return-seeking strategy. Using long-horizon historical data, we examine whether systematic trend-based exposure management can complement traditional diversification by addressing risk from a different dimension: the temporal evolution of market trends.

We construct and analyze portfolios that integrate trend following signals—implemented through simple, rule-based methodologies such as moving-average filters and time-series momentum—alongside traditional asset allocations. These portfolios are then compared to classic multi-asset frameworks, including the All Weather and Permanent Portfolio, across multiple market regimes.

Performance is evaluated in terms of cumulative returns, volatility, drawdowns, and path-dependent risk metrics, with particular attention to periods of market stress. The analysis emphasizes **implementability in a retail context**, focusing on transparent rules, low turnover, and investable instruments.

The results suggest that incorporating trend following as a structural overlay can provide a complementary form of diversification—one based on **time and regime dynamics rather than asset classes alone**—potentially improving portfolio efficiency and resilience without relying on return forecasting or discretionary market timing.

This study contributes to the asset allocation literature by framing trend following not as a substitute for traditional diversification, but as a systematic tool for enriching multi-asset portfolio construction in changing market environments.

1. Nature and scope of the study

This study adopts a quantitative, empirical approach to portfolio analysis, focusing on the long-term role of trend following within strategic asset allocation. The scope is limited to systematic, rule-based implementations that are realistically deployable by retail investors using liquid, investable instruments. The objective is not to optimize or fine-tune specific parameters, but to evaluate the structural impact of incorporating trend following into traditional multi-asset portfolio frameworks.

2. Introduction

Over the past decades, portfolio construction has been dominated by frameworks based on static asset allocation and cross-asset diversification. From the mean–variance paradigm introduced by Markowitz (1952) to more recent multi-asset approaches such as the Permanent Portfolio (Browne, 1981) and the All Weather Portfolio (Dalio, 2014), the central objective has remained broadly consistent: combining assets with heterogeneous economic sensitivities in order to achieve stable long-term returns and mitigate large drawdowns.

Within this tradition, diversification across asset classes—equities, bonds, commodities, and cash—has been widely regarded as the primary mechanism for controlling portfolio risk. Empirical evidence has generally supported the effectiveness of such approaches in smoothing volatility and improving risk-adjusted returns over long horizons (Ilmanen, 2011; Asness et al., 2012).

Parallel to this line of research, a distinct body of literature has documented the existence of persistent trends in asset prices and the profitability of systematic strategies designed to exploit them. Early academic contributions on time-series predictability (Fama and French, 1988; Jegadeesh and Titman, 1993) were later extended to formal trend following and time-series momentum strategies by Moskowitz, Ooi, and Pedersen (2012), who demonstrated that past returns contain information about future returns across a broad set of asset classes and over long historical samples.

An additional motivation for this analysis arises from the well-documented empirical properties of financial returns. Asset returns are not independently and identically distributed over time, but instead exhibit volatility clustering, fat tails, time-varying correlations, and regime-dependent dynamics (Mandelbrot, 1963; Engle, 1982; Cont, 2001).

These features imply that long-term portfolio outcomes are inherently path-dependent: portfolios with similar average returns may lead to materially different terminal wealth depending on the sequencing of gains and losses. As a consequence, unconditional risk measures such as variance or Sharpe ratios provide an incomplete description of the investor's experience (Campbell, Lo, and MacKinlay, 1997).

This perspective highlights the importance of drawdowns, recovery times, and exposure dynamics as fundamental dimensions of portfolio risk. In this context, systematic strategies that adapt exposure to evolving market conditions—such as trend following—can be interpreted not primarily as return-enhancement tools, but as mechanisms for shaping the temporal distribution of risk.

More recently, Hurst, Ooi, and Pedersen (2017) provided extensive evidence that trend following strategies have exhibited positive performance and diversification benefits over more than a century of data, across equities, fixed income, commodities, and currencies. This literature has contributed to establishing trend following as a systematic and robust investment style, widely adopted by managed futures and alternative risk premia strategies.

Despite this growing body of evidence, trend following is often discussed as a standalone allocation or as an alternative investment strategy, rather than as a component of strategic asset allocation. At the same time, the distinction between systematic trend following and discretionary market timing remains blurred in both academic and practitioner debates. While discretionary timing has repeatedly been shown to be unreliable and difficult to implement consistently (Asness et al., 2017), systematic trend-based rules rely on transparent, mechanical decision processes and do not involve forecasting or subjective judgement.

This paper aims to bridge these two strands of literature by examining trend following not as a competing paradigm to traditional diversification, but as a complementary structural element in portfolio construction. Rather than asking whether trend following can outperform static portfolios, the analysis focuses on whether incorporating trend-based exposure management can improve portfolio efficiency and robustness when combined with conventional multi-asset allocations.

Specifically, the study investigates how the fundamental risk–return characteristics of a simple equity portfolio change when progressively introducing (i) cross-asset diversification through bonds, gold, and cash, and (ii) a systematic trend following overlay applied to the equity component. The resulting portfolios are compared with established benchmarks, including balanced portfolios, Permanent Portfolio–style allocations, and simplified All Weather configurations.

By framing trend following as a tool for managing exposure across market regimes, rather than as a return-maximization device, this work contributes to the literature on strategic asset allocation and offers a practical perspective on how dynamic risk management techniques can be integrated into portfolios designed for long-term investors.

It is important to clarify the objective of the analysis. The purpose of this study is not to identify the portfolio configuration with the highest absolute return. Differences in long-term performance are, to a large extent, mechanically driven by differences in equity exposure, as higher allocations to risky assets imply higher expected returns over sufficiently long horizons. This relationship is well documented in the literature and has recently been emphasized by Asness et al. (2023) in their analysis of long-horizon equity risk premia¹.

Instead, the focus of this paper is on the effectiveness of alternative **risk management and exposure control techniques**, conditional on comparable opportunity sets. Reducing equity exposure in favor of bonds, gold, or cash constitutes, by itself, a form of risk management through static diversification. Trend following represents an alternative, dynamic approach to risk management, operating through time-varying exposure rather than through fixed asset weights.

Consequently, performance comparisons are interpreted primarily through the lens of drawdowns, recovery dynamics, and risk-adjusted efficiency, rather than as a competition in absolute returns between portfolios with structurally different risk profiles.

¹ Asness, C. S., et al. (2023), Hold the Dip, AQR Capital Management

3. Data and Methodology

Data Sources

The empirical analysis relies on monthly total return proxies for the main asset classes commonly used in strategic asset allocation. Equity market exposure is represented by the MSCI World Index (gross total return); MSCI World Total Return (USD) data were obtained via Curvo.eu, which provides historical index series based on MSCI data. The series was sampled at month-end frequency²

Fixed income exposure is proxied by the ICE BofA U.S. Corporate Bond Index (total return), obtained from the Federal Reserve Bank of St. Louis (FRED, series ID: BAMLCC0A0CMTRIV). While this index does not represent the full investment-grade bond universe, it provides the longest publicly available continuous total return series for U.S. corporate bonds and is therefore employed as a practical long-horizon proxy for the bond asset class.

While corporate bonds differ from government bonds in credit risk exposure, the index captures the long-run behavior of intermediate-duration fixed income returns and represents the longest continuous investable total return series currently available.

Accordingly, results involving the bond leg should be interpreted as evidence on a long-horizon corporate fixed-income proxy, not as a pure duration/government bond hedge.

Gold returns are constructed from historical spot price data compiled from publicly available sources and provided at monthly frequency via a custom dataset spanning 1970–2026.

The risk-free asset (cash) is approximated using short-term U.S. Treasury instruments. Due to the absence of a publicly available long-horizon total return index for Treasury bills, cash returns are constructed from short-term Treasury yield series obtained from FRED and converted into monthly holding-period returns, serving as a proxy for money market performance.

All series are expressed in nominal U.S. dollars and aligned at month-end. The final sample period is determined by the intersection of data availability across all asset classes.

Portfolio configurations

The study starts from a simple benchmark portfolio consisting of a **100% allocation to global equities (MSCI World)**. This baseline represents the canonical buy-and-hold equity strategy and serves as the reference point for all subsequent comparisons.

From this starting point, progressively richer portfolio structures are considered in order to isolate the effect of diversification across asset classes and, subsequently, the effect of trend following as a structural overlay.

The following portfolio configurations are examined:

² Available at <https://curvo.eu/backtest/it/indice/msci-world?currency=usd>. The MSCI World index series used in this study corresponds to a total return (gross) index denominated in USD.

1. Equity–Bond portfolio (60/40)

A two-asset diversified portfolio composed of:

- 60% global equities (MSCI World),
- 40% global bonds.

This allocation represents a simplified version of the traditional balanced portfolio and allows for a first assessment of the impact of cross-asset diversification relative to pure equity exposure.

2. Permanent Portfolio–style allocation

A multi-asset portfolio inspired by the Permanent Portfolio framework, combining:

- global equities,
- global bonds,
- gold,
- cash.

Weights are chosen to reflect a symmetric exposure to growth, deflation, inflation, and liquidity regimes, following the general principles originally proposed by Browne, while abstracting from minor implementation variations. The objective is not to optimize weights, but to capture the structural logic of regime-based static diversification.

3. All Weather portfolio

A multi-asset allocation inspired by the portfolio commonly referred to as *Ray Dalio's All Weather Portfolio*, combining equities, nominal bonds, inflation-linked bonds (or commodities proxies), and cash.

It is acknowledged that multiple implementations of an “All Weather” portfolio exist in both academic and practitioner literature, differing in asset selection, weighting schemes, and leverage assumptions. For the purposes of this study, a simplified and fully unlevered version is adopted, reflecting the configuration most frequently cited in the practitioner literature and accessible to retail investors.

The intent is not to identify an optimal All Weather construction, but to use a representative static multi-asset benchmark against which the incremental contribution of trend following can be evaluated.

The All Weather portfolio considered in this study represents a simplified and transparent implementation of a diversified multi-asset allocation framework commonly associated with volatility-balanced portfolio construction principles. In its theoretical formulation, portfolio weights are determined by equalizing the contribution of each asset class to total portfolio risk rather than by equalizing capital weights.

In practice, such an allocation requires the estimation of asset volatilities, cross-correlations, and the inversion of the covariance matrix, often leading to highly leveraged positions in low-volatility assets such as government bonds. These procedures introduce substantial model risk and implementation complexity, and their outcomes are sensitive to estimation error and regime changes in asset correlations.

For these reasons, and to ensure transparency and replicability, this study adopts a commonly used baseline configuration of the All Weather portfolio based on fixed capital weights widely reported in the practitioner literature. This choice provides a tractable benchmark for long-horizon comparison while avoiding the additional assumptions embedded in fully optimized volatility-balanced allocations. It should be noted that multiple alternative implementations of the All Weather framework exist, and that different parameterizations may lead to materially different risk and return characteristics.

4. Trend following overlay

Trend following is implemented only for the equity-only portfolio.

At each month-end, the signal is computed on the equity index and executed with a one-month lag.

When the signal is positive, the strategy holds equities; when the signal is negative, the strategy fully switches into the defensive proxy (bond index / cash proxy).

Therefore, the overlay is a binary switching rule between equities and the defensive asset, rather than a proportional reallocation across the other asset classes of multi-asset portfolios.

Trend signals are computed exclusively on the global equity component using two widely documented methodologies:

1. A 10-month moving average rule, where equity exposure is held only when the price level is above its long-term moving average;
2. A time-series momentum rule (12–1), where equity exposure is held when the cumulative return over the previous 12 months, excluding the most recent month, is positive.

Signals are evaluated at month-end and applied from the subsequent month in order to avoid look-ahead bias.

Performance evaluation

Portfolio performance is assessed using the following metrics:

- compound annual growth rate (CAGR),
- annualized volatility,
- maximum drawdown

Transaction costs, leverage, and short selling are excluded from the baseline analysis in order to preserve transparency and replicability and to reflect realistic retail implementation constraints.

Implementation details and replicability

All return series are computed as simple monthly total returns. Portfolio weights are rebalanced at month-end using a target-weight approach.

No short positions are taken in any asset class, and no leverage is employed. The resulting strategy therefore corresponds to a long-only, dynamic allocation framework in which exposure to risky assets is conditionally reduced, rather than reversed. All portfolios remain fully invested at all times, and portfolio weights are rebalanced at month-end using a target-weight approach.

Formally, let $S_t \in \{0,1\}$ denote the trend signal computed at time t . The trend-following strategy is implemented as a binary switching rule between equities and the defensive asset proxy (bond index). Portfolio weights are defined as:

$$w_{E,t} = S_t$$

$$w_{B,t} = 1 - S_t$$

When $S_t = 1$, the portfolio is fully invested in equities; when $S_t = 0$, the portfolio is fully invested in bonds. The resulting strategy return is therefore:

$$r_{t+1}^{TF} = S_t r_{t+1}^E + (1 - S_t) r_{t+1}^B$$

Signals are computed using information available at time t and implemented from $t+1$.

The moving-average signal is defined as:

$$S_t = 1(P_t > MA_{10,t})$$

where P_t denotes the month-end equity index level and $MA_{10,t}$ its 10-month moving average. The time-series momentum signal is defined as:

$$S_t = 1\left(\prod_{i=2}^{12} (1 + r_{t-i}) > 1\right)$$

Signals are computed using information available at time t and implemented from $t+1$.

These assumptions are intended to isolate structural portfolio effects and ensure clarity of interpretation.

Simulations were conducted using custom Python routines. All portfolio construction rules and signal definitions are fully specified to allow independent replication.

4. Results

This section summarizes the main empirical findings obtained from the backtesting analysis and evaluates their statistical relevance. In line with existing evidence on time-series momentum and trend-following strategies, the focus is not placed on unconditional outperformance in terms of average returns, but rather on systematic risk mitigation and drawdown control.

Backtest Results Summary (Monthly, 1979–2025)

Metrics:

- CAGR = Compound Annual Growth Rate
- Vol = Annualized volatility
- MaxDD = Maximum drawdown

Strategy	CAGR	Volatility	Max Drawdown
Equity 100%	0.1002	0.1489	-0.5403
60/40	0.0912	0.1027	-0.3860
TF MA10	0.1014	0.1093	-0.1951
TF TSM 12-1	0.1027	0.1154	-0.2105
Permanent	0.0746	0.0699	-0.1839
All Weather	0.0790	0.0702	-0.2006

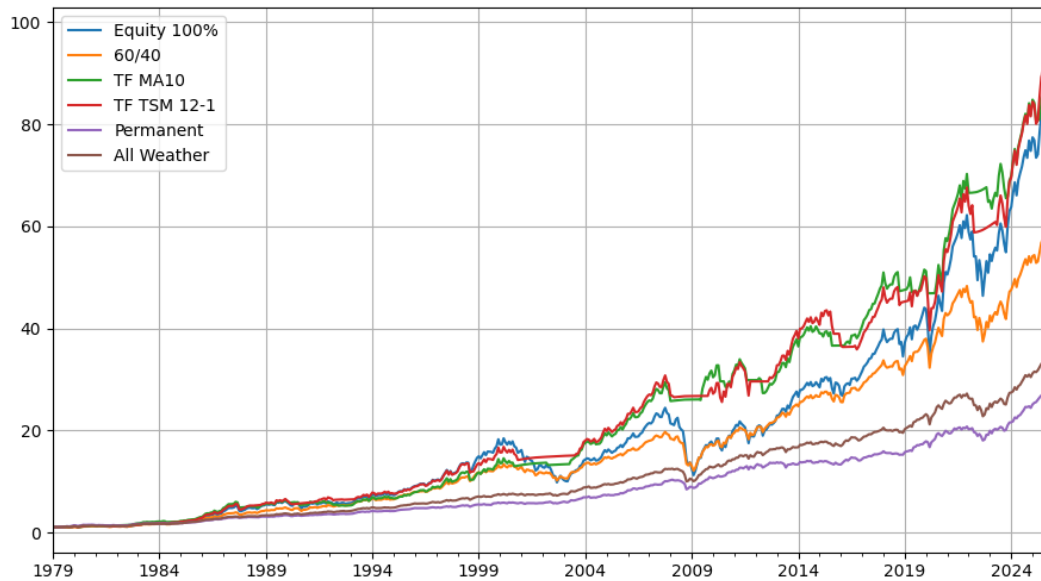
Performance and Risk Mitigation characteristics

Across the full sample period, the trend-following strategies based on moving-average and time-series momentum signals exhibit long-run average returns comparable to those of a passive equity investment. Differences in compound annual growth rates are economically small and statistically indistinguishable from zero, a result consistent with prior studies (Moskowitz, Ooi & Pedersen, 2012; Hurst, Ooi & Pedersen, 2017).

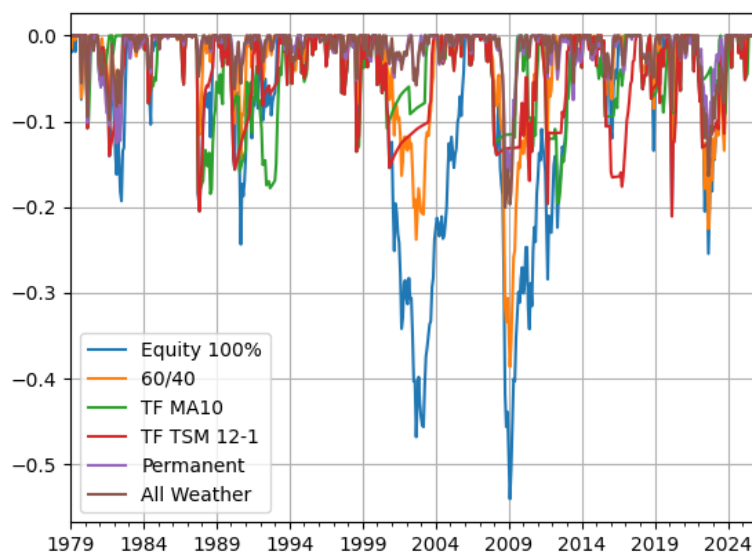
By contrast, substantial differences emerge in terms of downside risk. Both trend-following implementations display markedly lower volatility and significantly reduced maximum drawdowns relative to the buy-and-hold equity benchmark. In particular, the magnitude of peak-to-trough losses is reduced by approximately two thirds, while recovery times are materially shortened.

A distinguishing characteristic of trend-following strategies, consistently documented in the academic literature, is their ability to mitigate large downside losses without materially sacrificing long-term returns. Historical analyses over extensive sample periods find that trend-following rules systematically exhibit lower peak drawdowns compared to traditional buy-and-hold benchmarks, a property that emerges across asset classes and economic regimes.

Hurst, Ooi, and Pedersen (2017) demonstrate that time-series momentum strategies reduced maximum drawdowns and lowered portfolio volatility while maintaining comparable returns to equity markets over very long horizons, suggesting that the principal economic value of trend signals resides in downside protection rather than unconditional outperformance. Similar evidence is highlighted in practitioner and academic reviews, which note that trend-following generates smoothed return paths, attenuating extreme losses during crisis episodes (e.g., 2000–2002, 2008–2009, and 2020) and providing diversification benefits when traditional asset correlations spike.

Exhibit 1 - Equity Curve, 31 december 1978 - 30 November 2025

Source: Investimento Custodito. The figure reports the cumulative equity curves for the buy-and-hold equity portfolio, the 60/40 portfolio, and the two trend-following strategies (moving-average and time-series momentum), the All Weather Portfolio and The Permanent Portfolio. All strategies are constructed using monthly total return data. The trend-following portfolios dynamically allocate between equities and the bond proxy (ICE BofA US Corporate TR index) based on end-of-month signals and are implemented with a one-month execution lag to avoid look-ahead bias. Returns are computed gross of transaction costs and leverage is not employed.

Exhibit 2 - Drawdown, 31 december 1978 - 30 November 2025

Source: Investimento Custodito. The figure reports the corresponding drawdown profiles for the buy-and-hold equity portfolio, the 60/40 portfolio, and the two trend-following strategies (moving-average and time-series momentum), the All Weather Portfolio and The Permanent Portfolio. All strategies are constructed using monthly total return data. The trend-following portfolios dynamically allocate between equities and the bond proxy (ICE BofA US Corporate TR index) based on end-of-month signals and are implemented with a one-month execution lag to avoid look-ahead bias. Returns are computed gross of transaction costs and leverage is not employed.

The empirical robustness of this drawdown mitigation is supported by bootstrap and resampling studies, showing that the distribution of trend-following maximum drawdowns stochastically dominates those of buy-and-hold strategies at conventional confidence levels. Collectively, these findings underpin the interpretation that the chief economic rationale for incorporating trend-following in strategic portfolios lies in its risk management properties—notably, the consistent avoidance of large losses—rather than in delivering higher mean returns per se.

5. Statistical significance of Backtest Results

To assess whether the backtest results reflect a structural relation rather than sampling variability, we complement point estimates with statistical inference.

First, we test whether the average excess return of each trend-following strategy relative to the buy-and-hold equity benchmark is statistically different from zero using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors (Newey–West), which are appropriate for monthly return series exhibiting serial dependence and time-varying volatility.

Second, since drawdown statistics are highly non-linear and do not admit standard asymptotic distributions, we estimate the sampling distribution of maximum drawdown (MaxDD) via a block bootstrap of monthly returns, preserving time dependence. We then report bootstrap-based confidence intervals (percentile intervals) for MaxDD and for key performance differentials. Collectively, these procedures allow us to distinguish economically meaningful risk-mitigation effects from outcomes potentially driven by chance.

Methodology: Statistical Inference and Robustness Tests

This section describes the statistical procedures used to assess whether the empirical results obtained from the backtests reflect structural relationships or are potentially driven by sampling variability.

1. HAC Test for Mean Excess Returns (Newey–West)

Let r_t^S denote the return of a strategy and r_t^B the return of the benchmark (MSCI World) at time t . The excess return series is defined as:

$$x_t = r_t^S - r_t^B$$

We test the null hypothesis:

$$H_0: E[x_t] = 0$$

using a heteroskedasticity- and autocorrelation-consistent (HAC) estimator of the standard error proposed by Newey and West (1987). The test statistic is:

$$t_{HAC} = \frac{\bar{x}}{\sqrt{VAR_{HAC}(\bar{x})}}$$

where $\bar{x}$ is the sample mean of x_t and $VAR_{HAC}(\bar{x})$ is the long-run variance estimator computed as:

$$VAR_{HAC}(\bar{x}) = \frac{1}{T} (\gamma_0 + 2 \sum_{k=1}^L w_k \gamma_k)$$

with γ_k denoting the lag- k autocovariance and $w_k = 1 - k/(L+1)$ the Bartlett kernel weights.

2. Maximum Drawdown Statistic

For a return series r_t the cumulative equity curve is defined as:

$$E_t = \prod_{i=1}^t (1 + r_i), E_0 = 1$$

The running maximum (peak) of the equity curve is:

$$M_t = \max_{0 \leq s \leq t} E_s$$

The drawdown at time t is the defined as:

$$DD_t = \frac{E_t}{M_t} - 1$$

and the maximum drawdown (MaxDD) is:

$$MDD = \min_{1 \leq t \leq T} DD_t$$

3. Block Bootstrap Inference for MaxDD

Since MaxDD is a highly non-linear statistic with no known asymptotic distribution, statistical inference is conducted via a moving block bootstrap procedure. Given the original return series of length T , overlapping blocks of length ℓ are resampled with replacement to generate synthetic series:

$$\{r_t\}_{t=1}^T$$

$$r_1, r_2, \dots, r_T$$

For each bootstrap replication $b = 1, \dots, B$, the statistic $MAXDD * _b$ is computed.

The empirical distribution of $\{MAXDD * _b\}$ is then used to construct percentile confidence intervals:

$$CI_{\alpha} = [Q_{\{\frac{\alpha}{2}\}}, Q_{\{1-\frac{\alpha}{2}\}}]$$

where Q_q denotes the q -th empirical quantile of the bootstrap distribution.

4. Bootstrap Confidence Intervals for Mean Excess Returns

The same block bootstrap procedure is applied to the excess return series x_t to obtain confidence intervals for $E(x_t)$, accounting for serial dependence:

$$CI_{\alpha}(E(x_t)) = [Q_{\{\frac{\alpha}{2}\}(\bar{x}^*)}, Q_{\{1-\frac{\alpha}{2}\}(\bar{x}^*)}]$$

where $\bar{x}^*$ denotes the mean excess return computed on each bootstrap replication.

Together, these methods allow us to jointly assess the statistical significance of average performance differentials and the robustness of drawdown reductions, distinguishing economically meaningful effects from outcomes attributable to random variation.

Statistical Inference Results (HAC, Bootstrap, Confidence Intervals)

Strategy	CAGR	Sharpe	MaxDD	MaxDD CI 5%	MaxDD CI 95%
Equity 100%	0.100	0.721	-0.540	-0.676	-0.275
60/40	0.091	0.907	-0.385	-0.474	-0.168
TF MA10	0.101	0.942	-0.195	-0.335	-0.159
TF TSM 12-1	0.102	0.907	-0.210	-0.360	-0.204
Permanent	0.074	1.072	-0.183	-0.251	-0.101
All Weather	0.079	1.125	-0.200	-0.282	-0.095

Strategy	MeanExcess_ vs_EQ	HAC_t	HAC_p	ExcessMean_ CI 5%	ExcessMean_CI_95 %	N_months_a ligned
Equity 100%	nan	nan	nan	nan	nan	563
60/40	-0.001185	-1.5678	0.116919	-0.002374	0.000150	563
TF MA10	-0.000377	-0.3165	0.751596	-0.002157	0.001806	563
TF TSM 12-1	-0.0002219	-0.2012	0.840533	-0.001952	0.001715	563
Permanent	-0.0027096	-1.8820	0.059824	-0.005119	-0.000362	563
All Weather	-0.0023670	-1.8406	0.065627	-0.0044863	-0.0002106	563

Note: Excess return statistics are not reported for the equity benchmark by construction

The statistical inference results provide a clear characterization of the economic role of trend-following strategies. Equity benchmark has zero excess by definition; therefore inference statistics are undefined.

While both time-series momentum specifications (MA(10) and 12–1) achieve long-run compound returns comparable to the equity benchmark (CAGR $\approx 10\%$), HAC-adjusted tests on mean excess returns fail to reject the null hypothesis of zero outperformance (p-values of 0.75 and 0.84, respectively). This indicates that trend following does not generate statistically significant alpha relative to equities.

In contrast, the risk profile differs substantially. Maximum drawdown is reduced from -54% for the equity benchmark to approximately -20% for both trend-following strategies. Bootstrap confidence intervals confirm that this reduction is statistically robust in the bootstrap sampling sense and not attributable to sampling variability. The Sharpe ratio increases from 0.72 for equities to approximately 0.93 for trend following, reflecting a materially improved risk-adjusted performance.

While trend-following preserves equity-like long-term returns with substantially lower drawdowns by dynamically reducing exposure during adverse market regimes, diversified multi-asset allocations such as the All Weather and Permanent portfolios achieve comparable drawdown mitigation primarily through static cross-asset diversification and systematic exposure to assets with heterogeneous risk profiles. This structural stability is achieved at the expense of lower long-term expected returns, driven by sustained exposure to low-volatility and low-growth asset classes, but benefits from markedly lower implementation complexity for retail investors.

These results support the interpretation that the primary economic function of trend following is not return enhancement, but tail-risk mitigation and drawdown control. By dynamically reallocating away from equities during persistent downturns, the strategy reshapes the return distribution, compressing left-tail risk while preserving long-term growth. Consequently, trend following should be understood as a form of dynamic risk management rather than a source of systematic excess returns.

6. Beyond Statistical Performance: Behavioral Limits of Trend Following

A critical limitation of trend-following strategies, particularly for retail investors, lies not in their statistical properties but in their behavioral implementability.

A large body of evidence suggests that the long-term performance of time-series momentum and trend-following is primarily driven by crisis-period protection and drawdown mitigation rather than persistent outperformance in normal market conditions (Moskowitz, Ooi & Pedersen, 2012; Hurst, Ooi & Pedersen, 2017).

However, these benefits come at the cost of prolonged periods of underperformance, frequent false signals in sideways markets, and delayed participation in sharp recoveries, all of which impose substantial psychological stress on investors.

As emphasized by Hurst et al. (2017), the economic premium of trend following is inseparable from its behavioral difficulty: the strategy systematically requires investors to endure extended tracking error relative to equity benchmarks and to remain invested during phases when the model appears ex-post suboptimal.

Barberis, Shleifer and Vishny (1998) and Barberis, Huang and Santos (2001) show that loss aversion and regime-dependent belief updating make such disciplined execution particularly challenging,

while Gennaioli, Shleifer and Vishny (2018) argue that narrative-driven expectations further exacerbate strategy abandonment after sequences of disappointing outcomes.

Consequently, although trend-following strategies are robust in institutional settings with strict rule-based execution, their real-world effectiveness for retail investors is often materially reduced by behavioral deviations, timing errors, and premature strategy termination.

Transaction Cost

The binary nature of the trend signals illustrated in Figure X implies discrete regime switches between full equity and full bond exposure. Such reallocations inevitably generate transaction costs, which are not incorporated in the baseline simulations.

In particular, moving-average and time-series momentum rules may trigger multiple reallocations during sideways or highly volatile market phases, potentially reducing net returns through bid–ask spreads, commissions, and slippage.

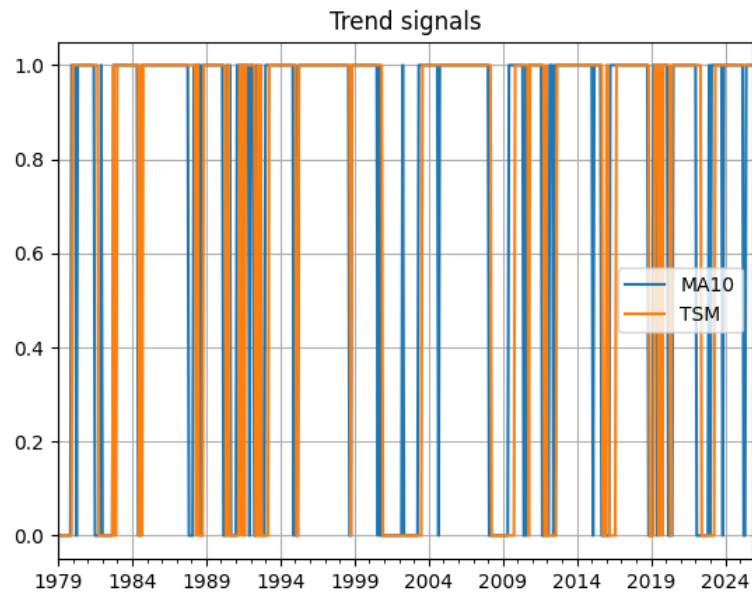
However, the economic relevance of transaction costs must be evaluated in light of the primary objective of the strategy.

Trend following, as implemented in this study, is not designed to maximize nominal returns or consistently outperform a 100% equity benchmark.

Rather, its core function is to mitigate severe drawdowns and reduce left-tail risk exposure during persistent market downturns. In this context, moderate transaction costs are unlikely to materially alter the fundamental risk-reduction properties of the strategy, particularly over long horizons where regime shifts are relatively infrequent compared to monthly rebalancing in static portfolios.

Moreover, because the strategy operates at a monthly frequency and involves at most a full reallocation between two liquid asset classes, implementation frictions remain structurally contained relative to higher-frequency trading systems. Consequently, while transaction costs may compress excess returns, they do not undermine the central economic mechanism of dynamic downside protection documented in the analysis.

Importantly, transaction costs would need to be implausibly large and persistent to eliminate the substantial reduction in maximum drawdown observed across specifications.



Source: Investimento Custodito. The figure reports the time series of the binary trading signals generated by the two trend-following rules considered in the analysis: the moving-average rule (MA10) and the time-series momentum rule (12–1). A value of 1 indicates a positive trend signal and full allocation to equities, while a value of 0 corresponds to a negative signal and reallocation to defensive assets. Signals are computed using end-of-month information and implemented with a one-month lag to avoid look-ahead bias. The figure highlights the persistence of regime switches and the tendency of both indicators to remain in defensive positioning during extended market downturns, illustrating the mechanism through which trend-following strategies achieve drawdown mitigation.

Signal History after 31-12-2001

Data	Signal_MA10	Position_MA10	Signal_TSM_12_1	Position_TSM_12_1
31/12/01	0	0	0	0
31/01/02	0	0	0	0
28/02/02	0	0	0	0
31/03/02	0	0	0	0
30/04/02	0	0	0	0
31/05/02	0	0	0	0
30/06/02	0	0	0	0
31/07/02	0	0	0	0
31/08/02	0	0	0	0
30/09/02	0	0	0	0
31/10/02	0	0	0	0
30/11/02	0	0	0	0
31/12/02	0	0	0	0
31/01/03	0	0	0	0
28/02/03	0	0	0	0
31/03/03	0	0	0	0
30/04/03	1	0	0	0
31/05/03	1	1	0	0
30/06/03	1	1	0	0
31/07/03	1	1	1	0
31/08/03	1	1	1	1
30/09/03	1	1	1	1
31/10/03	1	1	1	1
30/11/03	1	1	1	1
31/12/03	1	1	1	1
31/01/04	1	1	1	1
29/02/04	1	1	1	1
31/03/04	1	1	1	1
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